

Modul Praktikum

Pengembangan Aplikasi Basisdata

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MODUL PRAKTIKUM 10

Stored Procedure di SQL Server & Pemanggilan via ADO.NET

A. Tujuan

Setelah menyelesaikan praktikum ini mahasiswa mampu:

1. Memahami konsep Stored Procedure (SP)
2. Membuat SP untuk SELECT, INSERT, UPDATE, DELETE
3. Menggunakan parameter input dan output pada SP
4. Memanggil SP menggunakan ADO.NET
5. Mengintegrasikan SP dengan DataGridView dan BindingSource
6. Memahami kelebihan SP dibanding query biasa

B. Praktikum 1 (Membuat Stored Procedure)

LANGKAH 1 – Membuat Stored Procedure SELECT

Buka SQL Server Management Studio:

```
/* =====  
1. STORED PROCEDURE SELECT SEMUA DATA  
===== */  
  
CREATE PROCEDURE sp_GetMahasiswa  
AS  
BEGIN  
    SET NOCOUNT ON;  
  
    SELECT  
        NIM,  
        Nama,  
        JenisKelamin,  
        TanggalLahir,  
        Alamat,  
        KodeProdi,  
        TanggalDaftar  
    FROM Mahasiswa  
END
```

LANGKAH 2 – Stored Procedure dengan Parameter (SELECT by NIM)

```
/* =====  
2. STORED PROCEDURE SELECT BY NIM  
===== */  
  
CREATE PROCEDURE sp_GetMahasiswaByNIM  
    @NIM CHAR(11)  
AS  
BEGIN  
    SET NOCOUNT ON;  
  
    SELECT  
        NIM,  
        Nama,  
        JenisKelamin,  
        TanggalLahir,  
        Alamat,  
        KodeProdi,  
        TanggalDaftar  
    FROM Mahasiswa  
    WHERE NIM = @NIM  
END
```

LANGKAH 3 – Stored Procedure INSERT

```
/* =====  
3. STORED PROCEDURE INSERT  
===== */  
  
CREATE PROCEDURE sp_InsertMahasiswa  
    @NIM CHAR(11),  
    @Nama VARCHAR(100),  
    @JenisKelamin CHAR(1),  
    @TanggalLahir DATE,  
    @Alamat VARCHAR(200),  
    @KodeProdi CHAR(4),  
    @TanggalDaftar DATETIME  
AS  
BEGIN  
    SET NOCOUNT ON;  
  
    INSERT INTO Mahasiswa  
    (NIM, Nama, JenisKelamin, TanggalLahir, Alamat, KodeProdi, TanggalDaftar)  
    VALUES  
    (@NIM, @Nama, @JenisKelamin, @TanggalLahir, @Alamat, @KodeProdi, @TanggalDaftar)  
END
```

LANGKAH 4 – Stored Procedure UPDATE

```
/* =====  
4. STORED PROCEDURE UPDATE  
===== */  
  
CREATE PROCEDURE sp_UpdateMahasiswa  
    @NIM CHAR(11),  
    @Nama VARCHAR(100),  
    @JenisKelamin CHAR(1),  
    @TanggalLahir DATE,  
    @Alamat VARCHAR(200),  
    @KodeProdi CHAR(4)  
AS  
BEGIN  
    SET NOCOUNT ON;  
  
    UPDATE Mahasiswa  
    SET Nama = @Nama,  
        JenisKelamin = @JenisKelamin,  
        TanggalLahir = @TanggalLahir,  
        Alamat = @Alamat,  
        KodeProdi = @KodeProdi  
    WHERE NIM = @NIM  
END
```

LANGKAH 5 – Stored Procedure DELETE

```
/* =====  
5. STORED PROCEDURE DELETE  
===== */  
  
CREATE PROCEDURE sp_DeleteMahasiswa  
    @NIM CHAR(11)  
AS  
BEGIN  
    SET NOCOUNT ON;  
  
    DELETE FROM Mahasiswa  
    WHERE NIM = @NIM  
END
```

LANGKAH 6 – Stored Procedure COUNT (OUTPUT PARAMETER)

```
/* =====  
6. STORED PROCEDURE COUNT (OUTPUT PARAMETER)  
===== */  
  
CREATE PROCEDURE sp_CountMahasiswa  
    @Total INT OUTPUT  
AS  
BEGIN  
    SET NOCOUNT ON;  
    SELECT @Total = COUNT(*) FROM Mahasiswa  
END
```

C. Praktikum 2 (Implementasi di ado.net)

Hitung Total (OUTPUT Parameter)

```
private void HitungTotal()  
{  
    try  
    {  
        using (SqlConnection conn = new SqlConnection(connectionString))  
        {  
            using (SqlCommand cmd = new SqlCommand("sp_CountMahasiswa", conn))  
            {  
                cmd.CommandType = CommandType.StoredProcedure;  
  
                SqlParameter outputParam = new SqlParameter("@Total", SqlDbType.Int);  
                outputParam.Direction = ParameterDirection.Output;  
                cmd.Parameters.Add(outputParam);  
  
                conn.Open();  
                cmd.ExecuteNonQuery();  
  
                lblTotal.Text = "Total Mahasiswa: " + outputParam.Value.ToString();  
            }  
        }  
    }  
    catch (Exception ex)  
    {  
        MessageBox.Show("Gagal menghitung total: " + ex.Message);  
    }  
}
```

Load data (sp)

```
private void LoadData()  
{  
    using (SqlConnection conn = new SqlConnection(connectionString))  
    {  
        using (SqlCommand cmd = new SqlCommand("sp_GetMahasiswa", conn))  
        {  
            cmd.CommandType = CommandType.StoredProcedure;  
  
            using (SqlDataAdapter da = new SqlDataAdapter(cmd))  
            {  
                dtMahasiswa = new DataTable();  
                da.Fill(dtMahasiswa);  
  
                bindingSource.DataSource = dtMahasiswa;  
                dataGridView1.DataSource = bindingSource;  
  
                BindControls();  
            }  
        }  
    }  
  
    HitungTotal(); // ← tambahkan ini  
}
```

Insert

```
using (SqlCommand cmd = new SqlCommand("sp_InsertMahasiswa", conn))
{
    cmd.CommandType = CommandType.StoredProcedure;
    cmd.Parameters.AddWithValue("@NIM", txtNIM.Text);
    cmd.Parameters.AddWithValue("@Nama", txtNama.Text);
    cmd.Parameters.AddWithValue("@JenisKelamin", cmbJK.Text);
    cmd.Parameters.AddWithValue("@TanggalLahir", dtpTanggalLahir.Value.Date);
    cmd.Parameters.AddWithValue("@Alamat", txtAlamat.Text);
    cmd.Parameters.AddWithValue("@KodeProdi", txtKodeProdi.Text);
    cmd.Parameters.AddWithValue("@TanggalDaftar", DateTime.Now);

    conn.Open();
    cmd.ExecuteNonQuery();
}
```

Update

```
using (SqlCommand cmd = new SqlCommand("sp_UpdateMahasiswa", conn))
{
    cmd.CommandType = CommandType.StoredProcedure;

    cmd.Parameters.AddWithValue("@NIM", txtNIM.Text);
    cmd.Parameters.AddWithValue("@Nama", txtNama.Text);
    cmd.Parameters.AddWithValue("@JenisKelamin", cmbJK.Text); // + diperbaiki
    cmd.Parameters.AddWithValue("@TanggalLahir", dtpTanggalLahir.Value.Date);
    cmd.Parameters.AddWithValue("@Alamat", txtAlamat.Text);
    cmd.Parameters.AddWithValue("@KodeProdi", txtKodeProdi.Text);

    conn.Open();
    cmd.ExecuteNonQuery();
}
```

Delete

```
using (SqlCommand cmd = new SqlCommand("sp_DeleteMahasiswa", conn))
{
    cmd.CommandType = CommandType.StoredProcedure;

    cmd.Parameters.Add("@NIM", SqlDbType.Char, 11).Value = txtNIM.Text;

    conn.Open();
    int rowsAffected = cmd.ExecuteNonQuery();

    if (rowsAffected > 0)
        MessageBox.Show("Data berhasil dihapus");
    else
        MessageBox.Show("Data tidak ditemukan");
}
```

COUNT (OUTPUT)

```
private void HitungTotal()
{
    try
    {
        using (SqlConnection conn = new SqlConnection(connectionString))
        {
            using (SqlCommand cmd = new SqlCommand("sp_CountMahasiswa", conn))
            {
                cmd.CommandType = CommandType.StoredProcedure;

                SqlParameter outputParam = new SqlParameter("@Total", SqlDbType.Int);
                outputParam.Direction = ParameterDirection.Output;
                cmd.Parameters.Add(outputParam);

                conn.Open();
                cmd.ExecuteNonQuery();

                lblTotal.Text = "Total Mahasiswa: " + outputParam.Value.ToString();
            }
        }
    }
    catch (Exception ex)
    {
        MessageBox.Show("Gagal menghitung total: " + ex.Message);
    }
}
```